Annex II. Debt Sustainability Analysis

This annex conducts a Debt Sustainability Analysis (DSA), in accordance with the revised joint Bank-Fund debt sustainability framework (DSF) for low-income countries (LICs).[1] Haiti's risk of debt distress remains high and the likelihood of some risks has recently increased.[2] The country's main challenge is to undertake needed fiscal adjustment while protecting social and investment needs. Policies under the baseline scenario lead to a significant build up of external and public debt, which highlights the need to a swifter implementation of structural reforms. The country's external debt profile is particularly vulnerable to shocks on borrowing conditions and the exchange rate, while declines in the rate of growth have major negative impacts on public debt. In particular, a sudden stop of Petrocaribe inflows could jeopardize growth prospects and deteriorate the indicators of public debt.

Background

1. Haiti's external debt structure has undergone significant changes in recent years and borrowing terms have improved. The share of debt owed to traditional development partners has dropped owing to debt relief, while the share of non-Paris Club bilateral debt has increased reflecting continued disbursements from Venezuela. After benefitting from HIPC and MDRI, Haiti received additional debt relief following the earthquake of 2010, including from the IMF (US$268 million)[3], the World Bank (US$36 million), the IDB (US$486 million), the IFAD (that established a debt relief account with sufficient funds to service debt through 2020), Venezuela (US$395 million), and Taiwan Province of China (that waived interest payments and postponed principal repayments by five years). Following the delivery of debt relief, Haiti's debt stock declined substantially, as major bilateral development partners and multilateral development banks have all shifted to providing assistance on a grants-only basis. Major new disbursements are those related to the concessional Petrocaribe agreement with Venezuela, which have recently led to a gradual increase in the debt stock.

2. At end-2013, Haiti's stock of public sector debt amounted to US$1,627 million, (19.5 percent of GDP), composed almost exclusively by external debt on concessional

[1] World Bank and IMF (2009). "Review of Some Aspects of the Low-Income Country Debt Sustainability Framework." (IDA/SecM2009-0397; SM/09/216; BUFF/09/146); World Bank and IMF (2012), "Revisiting the Debt Sustainability Framework for Low Income Countries," (SM/12/10). This full DSA was done jointly by IMF and World Bank staffs.

[2] Thresholds are set according to the country's CPIA ranking. Haiti is classified as a weak performer based on its three-year 2010-12 average score of 2.90 in the World Bank's Country Policy and Institutional Assessment (CPIA) framework. For weak performers (defined as those with three-year average CPIA ratings below 3.25), the indicative thresholds for external debt sustainability are PV debt-to-GDP ratio of 30 percent and 27 percent when remittances are included; PV debt-to-exports ratio of 100 percent and 80 percent when remittances are included, PV debt-to-revenue ratio of 200 percent, debt service-to-exports ratio of 15 percent and 12 percent with remittances; and debt service-to-revenue ratio of 25 percent.

[3] Established in June 2010, the Post-Catastrophe Debt Relief (PCDR) Trust allows the Fund to provide debt relief when LICs are hit by catastrophic natural disasters.

terms (Text Table 1). External debt amounted to US$1,475 million, of which US$136 million corresponded to multilateral debt, and US$1,339 million to bilateral debt, mainly to Venezuela (US$1,235 million of Petrocaribe debt). Domestic debt was about US$152 million, mostly in the form of treasury bills largely held by commercial banks and about US$45 million is a publicly guaranteed debt to a commercial bank contracted by the state-owned electricity company EDH (*Electricité d'Haïti*).

3. Other outstanding creditor and debtor positions with the private sector appear in favor of the public sector. The non-financial public sector remains a net creditor to the consolidated banking system, mainly reflecting Petrocaribe deposits with commercial banks (about US$200 million at end-2013). In addition, the public sector maintains a debtor position *vis-à-vis* independent power producers (IPPs), for arrears accumulated by EDH for about US$96 million; in turn the IPPs have accumulating arrears to the central government of US$80 million, for the provision of fuel.

Text Table 1. Haiti: Structure of External Public Debt at end-2013

		in percent of	
	US$ millions	total debt	GDP 1/
Total	**1474.8**	**100.0**	**17.4**
Multilateral creditors	**135.9**	**9.2**	**1.6**
IMF	57.8	3.9	0.7
World Bank	0.0	0.0	0.0
IFAD	64.5	4.4	0.8
IDB	0.1	0.0	0.0
OPEC	13.5	0.9	0.2
Official bilateral creditors	**1338.9**	**90.8**	**15.8**
Venezeula	1249.1	84.7	14.8
PetroCaribe	1235.0	83.7	14.6
BANDES	14.1	1.0	0.2
Taiwan, Province of China	89.8	6.1	1.1

Sources: Haitian authorities; and Fund staff estimates.

1/ The debt ratio differs slightly from that in Tables 1a and 3a given the use of average, instead of end-of-period, exchange rates.

A. Macroeconomic Outlook, 2014-2034

4. Haiti's main challenge is to balance the need for a gradual fiscal adjustment with large social and investment needs. A stronger than warranted fiscal adjustment may result in a better debt profile at the cost of lower GDP growth and slower progress in addressing social needs. Conversely, too large a primary deficit could compromise macroeconomic stability. In particular, the baseline scenario is based on the following assumptions:

- Projections include an annual average real GDP growth of about 3.5 percent, starting from 4 percent growth rate in 2014 and then declining progressively to 3 percent by 2034.
- A stable real exchange rate vis-à-vis the US dollar.
- Fiscal projections include a gradual increase in domestic revenues of the central government to 18 percent of GDP by 2034 (from nearly 13 percent in 2013), while foreign

grants will gradually decrease to about 2 percent of GDP (from 8 percent in 2013). Primary expenditure of the central government is projected to decline slowly from 27 percent of GDP in 2013 to about 22 percent in 2034, with capital expenditure (domestically and externally financed) also decreasing to stabilize at 10-11 percent of GDP by 2034. This would involve an adjustment in the primary balance of the central government of about 4.5 percentage points of GDP during the projection period, with the primary deficit decreasing from more than 6 percent of GDP in 2013 to nearly 2 percent in 2034. In turn, the deficit of EDH is assumed to gradually decrease from nearly 1 percent of GDP in 2013 to balance by 2022. The primary deficit in 2034 (about 2 percent of GDP) would be consistent with debt stabilization. This suggests that provided the value of other parameters is maintained (namely GDP growth of 5 percent –in US$ terms–and an average nominal interest rate of about 1.8 percent –in US$ terms– prevailing in 2034), the debt ratio would begin decreasing if fiscal consolidation continued beyond 2034, a reasonable assumption.

- For the external outlook, projections consider a gradually improving trade deficit (from 34 percent of GDP in 2013 to 16 percent in 2034), and gradually increasing FDI (from 1.4 percent of GDP in 2013 to 4.5 percent by 2034). The projections further assume Petrocaribe debt inflows that remain fairly steady in nominal terms but fall as a share of GDP (from 4.7 percent of GDP to 1.6 percent by 2034), at prevailing financing concessional terms. As the country develops, transfers (both official and remittances) would decline gradually, from 27 percent of GDP in 2013 to 11 percent in 2034.

5. Gross financing needs will be covered by a combination of external and domestic debt. Gross financing needs are projected to average about 8.5 percent of GDP per year, which will be financed by external debt flows (about 5.5 percent of GDP) and by domestic debt (about 3 percent of GDP). The DSA assumes that external borrowing would be composed by Petrocaribe flows (2.5 percent of GDP on average) and additional external bilateral borrowing at concessional terms (3.2 percent of GDP on average). Domestic borrowing would comprise only treasury bills with 1-5 year maturities, and 5 percent interest rate (in US$ terms). For 2014-16, withdrawals from the PCDR account are also projected, for a total amount of 2.5 percent of GDP.[4]

[4] The baseline scenario assumes that the country will be able to secure the concessional financing needed to smooth out the adjustment of the primary deficit over several years, and that structural reforms proceed, but slowly.

Text Table 2. DSA 2014 vs. DSA 2012

	2012		Average 2013-17		Average 2018-31	
	Previous DSA	Current DSA	Previous DSA	Current DSA	Previous DSA	Current DSA
(annual percentage change, unless otherwise indicated)						
Real GDP	7.8	2.8	5.9	4.1	4.5	3.5
Consumer prices (period average)	7.7	6.8	4.6	5.3	3.0	5.0
(in percent of GDP, unless otherwise indicated)						
Total revenue and grants	29.5	25.3	24.0	20.6	19.8	20.7
Of which: Revenue	13.6	12.8	14.6	13.4	16.6	16.0
Primary expenditure	36.9	28.0	28.5	26.6	22.4	24.2
Of which: Capital expenditure	25.9	16.3	18.5	14.6	12.2	11.5
Overall balance	-7.7	-5.1	-4.9	-6.5	-3.4	-4.5
Current account balance	-4.5	-5.4	-4.5	-5.9	-4.2	-5.0
Exports of goods and services	15.0	16.9	14.6	18.5	14.1	21.3
Imports of goods and services	-54.6	-53.2	-43.3	-49.8	-32.0	-44.6

Source: Haitian authorities; and Fund staff estimates and projections.

6. The DSA projects that public debt will grow from 19.5 percent of GDP in 2013 to about 63 percent of GDP in 2034. In turn, external debt will grow from nearly 18 percent of GDP in 2013 to 53 percent of GDP in 2034. External debt would grow fairly rapidly in the near term but the pace of accumulation would fall to around 2 percent of GDP per year by the end of the projection period; the grant element of new borrowing will also decline gradually but will remain above 35 percent (Tables 1 to 3).

7. The main differences in the medium-term macroeconomic assumptions with respect to the previous DSA are as follow (Text Table 2),

- GDP is assumed to grow at a slower pace (3.5 percent) as opposed to 4.5–5 percent in the previous DSA, on account of the country's historical growth trend and international experience (See Box 1 on growth projections). Inflation would be slightly higher.
- Government revenue increases would be more gradual on account of the recent performance in revenue collection, and capital spending levels in the short-term would be more moderate but would remain sustained.
- Exports would be higher given recent trends and would expand in the long-term in part due to the HELP and HOPE initiatives, which provide advantageous conditions for apparel exports to the U.S. Imports would also remain at higher levels for all the projection period reflecting the country's needs to rebuild infrastructure, as well as Haiti's status as a net oil and food importer.
- Other major differences relate to the use of a new discount rate (5 percent) to calculate the grant element, and the inclusion of remittances in the analysis; the latter stood at 21 percent of GDP (106 percent of exports of goods and services) in 2013 and are projected to decline to 10 percent of GDP in 2034.[5]

[5]See Board paper "Unification of Discount Rates Used in External Debt Analysis for Low-Income Countries," (SM/13/271).

B. Debt Sustainability Outlook, 2014-34

8. The risk of external debt distress remains high under the baseline scenario. Three thresholds for external debt would be breached starting from 2021-22, including the ratios of PV of debt to GDP, to exports and to revenue (Figure 1). The ratio of debt service to revenue would not breach the threshold but would reach the 10 percent band around the threshold at the end of the projection period. While the ratio of debt service to exports would remain below the threshold during the projection period, it would be on an upward trend.

9. Haiti's external position is vulnerable to changes in borrowing terms and in the exchange rate. An increase by 200bps in the interest rate over the entire projection period would cause the thresholds for the PV to exports, and the debt service ratios to exports and to revenue to be breached very rapidly. In case of a 30 percent nominal depreciation of the gourde occurred in 2015, the PV of debt-to-GDP and to revenue would breach the associated thresholds as early as 2017-18.

10. Public debt indicators would deteriorate over time and show sensitivity to growth shocks (Figure 2). The ratio of the PV of public debt to GDP would reach 46 percent by 2034. In turn, the ratio of the PV of public debt-to-revenue would be on an increasing trend, reaching 206 percent at the end of the projection period, while debt service-to-revenue ratio would reach 33 percent by 2034. All debt indicators would increase to very high levels in response to shocks to growth and to the primary balance.

C. A 'Sudden Stop' of Petrocaribe Inflows

11. A sudden stop of Petrocaribe flows would lead, in the short-term, to a substantial fiscal adjustment and a decrease in GDP growth. This scenario assumes a sudden stop of Petrocaribe flows starting in FY2014/15, which would severely impact investment spending and the financing of the electricity sector. In addition, the shock to growth would impact domestic tax revenues, which would decrease below the baseline. The impact on the economy would be cushioned by a decrease in domestic revenues, a drawdown of Petrocaribe deposits in 2015-16 (of about 2.2 percent of GDP) and international reserves, and assumed additional bilateral external borrowing (on concessional terms). In particular, the deficit of EDH would likely decrease, given the lack of financing, through a combination of higher tariffs, longer blackouts, and some improvement in collection.

12. The lower concessional flows will result in lower investment and higher debt costs, which will constrain GDP growth. The decrease in investment and the temporary shortages in the provision of electricity would cause GDP growth to fall sharply in 2015-16 (by 3 and 2.5 percent points below the baseline respectively), and to remain lower than the baseline over the medium term. With lower GDP growth, tax collection would also decline on impact, to recover gradually thereafter.

Text Table 3. Sudden-Stop of Venezuela Flows

(in percent of GDP, unless otherwise noted)

	2014	2015	2016	2017	2018	2019
Baseline						
Real GDP growth (percent)	4.0	4.0	4.0	4.0	4.0	4.0
Public Debt	22.9	28.2	33.1	38.0	42.3	45.9
Primary Deficit NFPS	7.1	6.8	6.5	6.1	5.7	5.2
Central Government Revenue (excl. grants)	13.2	13.5	13.8	14.1	14.4	14.7
Central Government Primary Spending	26.9	26.6	26.4	26.1	25.8	25.6
of which, Transfers to Electricity Sector	1.4	1.3	1.2	1.1	1.0	1.0
of which, Public Investment	14.9	14.4	14.1	13.9	13.5	13.0
Alternative Scenario: Sudden Stop						
Real GDP growth (percent)	4.0	1.0	1.5	3.0	3.5	3.5
Public Debt	22.9	26.7	30.5	34.5	37.5	40.4
Primary Deficit NFPS	7.1	5.3	5.0	4.5	4.1	4.1
Central Government Revenue (excl. grants)	13.2	12.8	13.0	13.3	13.6	13.8
Central Government Primary Spending	26.9	24.6	24.4	24.1	23.8	23.9
of which, Transfers to Electricity Sector	1.4	0.8	0.7	0.6	0.5	0.5
of which, Public Investment	14.9	12.9	12.6	12.4	12.0	11.8

Source: IMF staff estimates and projections

13. The stock of public debt would remain broadly unchanged, but its composition would change in favor of less concessional debt. By 2034, total public debt would reach about 63 percent of GDP (similar than in the baseline). However, it is assumed that external debt would be about 13 percentage points of GDP lower than in the baseline (as Petrocaribe flows would not be substituted 1-to-1 by other concessional debt), while domestic debt would be higher. Importantly, without a strong deepening of the domestic financial system, such increase in domestic debt may crowd out credit to the private sector, further constraining growth.

14. Under this scenario the risk of external debt distress would remain high. The ratios of PV of debt to exports and to revenue would breach their respective thresholds by the end of the projection period. In four cases out of five the thresholds would be breached in case of sharp depreciations and changes in the borrowing terms. Public debt indicators would be somehow worse than in the baseline given the higher content of non concessional domestic borrowing. In particular, the ratio of the PV of public debt to GDP would reach 59 percent by 2034 (as opposed to 54 percent under the baseline); the ratio of the PV of public debt to revenue and the debt service would also deteriorate significantly with respect to the baseline.

D. Stronger Policies, Lower Vulnerabilities

15. Swift progress in the implementation of reforms should result in a stronger fiscal position and in an environment more conducive to growth. This scenario assumes that (i) the deficit of the state-owned electricity company is reduced faster than in the baseline; (ii) the effectiveness and efficiency of public investment improves *vis-à-vis* the baseline; and (iii) domestic revenues are higher than in the baseline, in part due to the elimination of the fuel price freeze. This results in a decrease in the primary deficit of the NFPS with respect to the baseline of about 1.5 percent of GDP by 2017. The increased effectiveness of public investment and the better performance in the electricity sector are assumed to have a positive impact of GDP growth, which is assumed to increase by 0.2 percentage points through the medium term.

Text Table 4. Stronger Policies, Lower Vulnerabilities

(in percent of GDP, unless otherwise noted)

	2014	2015	2016	2017	2018	2019
Baseline						
Real GDP growth (percent)	4.0	4.0	4.0	4.0	4.0	4.0
Public Debt	22.9	28.2	33.1	38.0	42.3	45.9
Primary Deficit NFPS	7.1	6.8	6.5	6.1	5.7	5.2
Central Government Revenue (excl. grants)	13.2	13.5	13.8	14.1	14.4	14.7
Central Government Primary Spending	26.9	26.6	26.4	26.1	25.8	25.6
of which, Transfers to Electricity Sector	1.4	1.3	1.2	1.1	1.0	1.0
of which, Public Investment	14.9	14.4	14.1	13.9	13.5	13.0
Active Scenario						
Real GDP growth (percent)	4.0	4.0	4.2	4.2	4.2	4.2
Public Debt	22.9	27.6	31.2	34.2	36.8	38.9
Primary Deficit NFPS	7.1	6.2	5.3	4.2	3.9	3.6
Central Government Revenue (excl. grants)	13.2	13.6	14.0	14.4	14.7	15.0
Central Government Primary Spending	26.9	26.3	25.6	25.0	24.8	24.6
of which, Transfers to Electricity Sector	1.4	1.0	0.5	0.0	0.0	0.0
of which, Public Investment	14.9	13.9	13.6	13.4	13.0	12.5

Source: IMF staff estimates and projections

16. The stock of public debt would decrease considerably *vis-à-vis* the baseline scenario. By 2034, total public debt would reach nearly 43 percent of GDP (against 63 percent in the baseline). In turn, domestic debt would reach only 7 percent of GDP, implying better financing terms, on average, than in the baseline. Credit to the private sector would not be crowd out, lifting a possible constraint to growth.

17. The risk of debt distress in this scenario would be reduced and vulnerabilities contained. Both external and public debt would peak by 2025 and would decrease thereafter, with no thresholds breached by the end of the projection period.

E. Conclusions

18. The updated DSA suggests that Haiti's risk of debt distress remains high, and that the likelihood of some risks has recently increased. The debt is also vulnerable to shocks to growth, borrowing terms and the exchange rate. A sudden stop of Petrocaribe financing would jeopardize growth prospects and reduce the grant element of new borrowing, thereby worsening the indicators of public debt. A stronger and faster implementation of structural reforms (including improving the performance the electricity sector, the effectiveness of public investment, and the focusing of subsidies), would result in lower debt accumulation through the medium term, and in the containment of vulnerabilities. It is therefore crucial that Haiti maintains prudent macroeconomic policies; strengthens the effectiveness of public investment, debt management, and PFM more generally, improves the sustainability of the electricity sector, and continues the implementation of structural reforms to improve the investment climate, in order to boost exports and growth.

19. The authorities broadly concurred with the main findings of the DSA. Staff and the authorities discussed the main assumptions and conclusions of the updated DSA, as well as main risks affecting surrounding the baseline. The authorities agreed with the need to increase the yield on growth of public investment, and more generally to contain the primary deficit of the non-financial public sector. They agreed on the need to speed up the implementation of reforms, but highlighted that a number of constrains have slowed down the process.[6]

[6] Discussions were held on a Seminar on Debt Sustainability that was co-hosted by IMF staff and the Haitian authorities in Port-au-Prince, during January 2014.

Box 1. Per Capita GDP Growth Projections

Haiti's growth performance during the past two decades has been poor. In per capita terms, GDP growth in the twenty years through 2013 was actually negative, of about -0.9 percent. This situates Haiti's performance in the bottom 5th percentile of comparator groups, including countries of all levels of development, low-income countries (LICs), and non-resource rich LICs (See Table).[1] Political instability exacerbated problems with growth, but average per capita GDP growth was negative even excluding crisis years.

Average Per Capita GDP Growth, %, 1993-2012

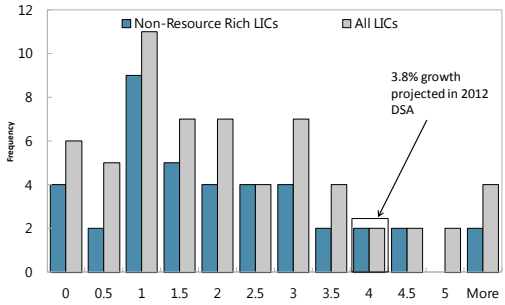
By percentile (number of observations)

Percentile	All Countries (164)	LICs (61)	Non-Resource Rich LICs (40)
Memo: Haiti 1993-2012			*-0.9*
5%	-0.2	-0.8	-0.8
10%	0.4	0.3	0.2
25%	1.1	0.7	0.7
50%	1.9	1.5	1.5
Memo: Haiti Proj. 2014-2034			*2.3*
75%	3.1	2.3	2.6
90%	4.4	4.4	3.9
95%	5.6	5.6	4.5

Data for 164 countries suggest that the median per capita GDP growth for all comparator groups fell in the 1.5 – 2.0 percent range during the past twenty years. While the median per capita GDP growth for all countries was 1.9 percent per year, the median rate was 1.5 percent for both LICs and non-resource rich LICs.

Per Capita Growth Performance Across LICs

(20-year average per capita growth rates)



Sources: IMF staff estimates.

This DSA assumes that Haiti will be able to sustain positive and significant per capita growth through the medium term. Concretely, average per-capita GDP growth for 2014-2034 is assumed at 2.3 percent, i.e., somewhat above the median growth observed for non-resource rich LICs during the past twenty years, but close to the median for all country groups. This would be the result of a projected average real GDP growth rate of 3.5 percent and population growth of 1.2 percent.[2] Although Haiti's achieving such growth rates would constitute a major shift from the past, the current growth baseline is nonetheless more conservative than that included in previous DSAs.[3]

This significant shift with respect to historical experience would be possible provided political and macroeconomic stability are maintained, inclusive institutions are strengthened, and the business climate is improved. Given the large space for catch-up growth, positive institutional change could generate large increases in total factor productivity, which combined with the increases projected in other factors, could easily generate growth rates of the magnitudes assumed. In this regard, baseline GDP growth rates are the result of expected population growth and increases in the capital stock (mainly on the back of relatively large levels of public investment), and low (but positive) contribution of total factor productivity. Stronger progress in structural reforms could yield larger increases in TFP (Section D). Relative large increases in TFP have been observed in LICs countries that significantly improve their policy performance.

Box 1. Per Capita GDP Growth Projections (concluded)

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[1] Data on per capita growth rates for 164 countries (including 61 LICs and 40 non-resource rich LICs) were obtained from the World Development Indicators database. The distinction between all LICs and non-resource rich LICs is based on "Macroeconomic Policy Frameworks for Resource-Rich Developing Countries" (IMF, 2012)

[2] Population projections are those of IHSI and United Nations.

[3] The previous DSA (2012) assumed an average per capita GDP growth rate of 3.8 percent per year over 2012-2032.

Table 1a. Haiti: Public Sector Debt Sustainability Framework, Baseline Scenario, 2011-2034

(In percent of GDP, unless otherwise indicated)

	Actual					Estimate	Projections								
	2011	2012	2013	Average 6/	Standard Deviation 6/	2014	2015	2016	2017	2018	2019	2014-19 Average	2024	2034	2020-34 Average
Public sector debt 1/	8.9	14.4	19.5			22.9	28.2	33.1	38.0	42.3	45.9		57.4	63.4	
o/w foreign-currency denominated	8.9	13.8	17.7			20.1	24.6	28.7	32.3	35.7	38.6		48.8	53.3	
Change in public sector debt	-4.2	5.5	5.1			3.4	5.3	4.9	4.9	4.3	3.6		1.6	0.0	
Identified debt-creating flows	3.6	1.4	5.4			5.4	5.2	4.7	4.3	3.7	3.2		1.6	-0.1	
Primary deficit	4.1	2.7	6.3	2.4	1.8	6.3	6.1	5.9	5.6	5.3	4.9	5.7	3.5	1.9	3.0
Revenue and grants	15.8	25.3	20.8			20.6	20.5	20.5	20.5	20.6	20.7		20.8	20.2	
of which: grants	5.9	12.5	8.1			7.4	7.0	6.7	6.4	6.2	6.0		4.8	2.0	
Primary (noninterest) expenditure	19.8	28.0	27.1			26.9	26.6	26.4	26.1	25.8	25.6		24.3	22.0	
Automatic debt dynamics	-0.5	-1.3	-0.9			-0.9	-1.0	-1.2	-1.4	-1.6	-1.7		-2.0	-1.9	
Contribution from interest rate/growth differential	-0.5	13.1	-0.7			-0.9	-1.0	-1.2	-1.4	-1.6	-1.7		-2.0	-1.9	
of which: contribution from average real interest rate	-0.1	13.4	-0.1			-0.2	-0.1	-0.1	-0.1	-0.1	-0.1		-0.1	-0.1	
of which: contribution from real GDP growth	-0.4	-0.2	-0.6			-0.8	-0.9	-1.1	-1.3	-1.5	-1.6		-1.9	-1.8	
Contribution from real exchange rate depreciation	0.0	-14.5	-0.2			0.0	0.0	0.0	0.0	0.0	0.0		...	...	
Other identified debt-creating flows	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Privatization receipts (negative)	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Recognition of implicit or contingent liabilities	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Debt relief (HIPC and other)	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Other (specify, e.g. bank recapitalization)	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Residual, including asset changes 2/	-7.7	4.1	-0.2			-2.0	0.1	0.2	0.6	0.5	0.4		0.1	0.1	
Other Sustainability Indicators															
PV of public sector debt	...	...	12.4			15.5	19.2	22.7	26.4	29.5	32.2		40.8	46.1	
o/w foreign-currency denominated	...	...	10.6			12.7	15.6	18.3	20.7	22.9	24.9		32.1	36.0	
o/w external	...	...	10.6			12.7	15.6	18.3	20.7	22.9	24.9		32.1	36.0	
PV of contingent liabilities (not included in public sector debt)	...	...	...			...	...	...	...	...	...		...	...	
Gross financing need 3/	4.1	3.2	8.1			9.1	12.2	12.0	13.6	14.9	15.1		17.0	18.5	
PV of public sector debt-to-revenue and grants ratio (in percent)	...	...	59.8			75.3	93.8	110.9	128.8	143.6	155.7		196.3	228.9	
PV of public sector debt-to-revenue ratio (in percent)	...	...	97.9			117.5	142.4	165.0	187.5	205.6	219.4		255.9	254.1	
o/w external 4/	...	...	83.5			96.4	115.9	132.7	147.1	159.5	169.8		201.6	198.5	
Debt service-to-revenue and grants ratio (in percent) 5/	0.2	1.8	2.6			2.4	13.6	9.6	12.3	15.4	14.8		23.1	31.6	
Debt service-to-revenue ratio (in percent) 6/	0.3	3.6	4.2			3.7	20.6	14.2	17.9	22.0	20.9		30.1	35.1	
Primary deficit that stabilizes the debt-to-GDP ratio	8.2	-2.8	1.1			2.9	0.8	1.0	0.7	1.0	1.3		1.9	1.9	
Key macroeconomic and fiscal assumptions															
Real GDP growth (in percent)	3.1	2.9	4.3	1.5	2.3	4.0	4.0	4.0	4.0	4.0	4.0	4.0	3.5	3.0	3.3
Average nominal interest rate on forex debt (in percent)	0.5	0.8	0.9	1.1	0.7	1.0	1.3	1.3	1.2	1.3	1.3	1.2	1.3	1.3	1.3
Average real interest rate on domestic debt (in percent)	...	...	...	...	...	-0.1	1.6	1.8	1.9	2.1	2.3	1.6	2.6	2.7	2.6
Real exchange rate depreciation (in percent, + indicates depreciation)	-0.3	-65.7	-1.7	-10.0	22.9	0.1	...	...	...	...	...	...	...	...	...
Inflation rate (GDP deflator, in percent)	3.7	19.4	6.6	14.1	7.3	5.1	5.0	5.0	5.0	5.0	5.0	5.0	5.0	5.0	5.0
Growth of real primary spending (deflated by GDP deflator, in percent)	28.5	14.5	1.5	11.7	16.0	3.3	3.0	3.0	3.0	3.0	3.0	3.0	2.5	2.0	2.3
Grant element of new external borrowing (in percent)	...	...	0.0	0.0	...	37.5	36.7	36.5	36.4	36.3	36.2	36.6	36.1	35.9	...

Sources: Haitian authorities; and staff estimates and projections.

1/ Public sector debt includes gross debt of the central government, the central bank and nonfinancial public sector institutions.

2/ Includes the primary deficit of nonfinancial public sector institutions (other than the central government), essentially the state-owned electricity company EDH.

3/ Gross financing need is defined as the primary deficit plus debt service plus the stock of short-term debt at the end of the last period.

4/ Revenues excluding grants.

5/ Debt service is defined as the sum of interest and amortization of medium and long-term debt.

6/ Historical averages and standard deviations are generally derived over the past 10 years, subject to data availability.

Table 2. Haiti: Sensitivity Analysis for Key Indicators of Public Debt 2014-2034

	Projections							
	2014	2015	2016	2017	2018	2019	2024	2034
PV of Debt-to-GDP Ratio								
Baseline	16	19	23	26	30	32	41	46
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	18	19	21	23	24	26	31	40
A2. Primary balance is unchanged from 2014	17	22	26	31	35	39	57	86
A3. Permanently lower GDP growth 1/	17	24	30	36	42	47	70	105
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviations in 2015-20	17	25	34	41	48	54	79	111
B2. Primary balance is at historical average minus one standard deviations in 2015-201	17	19	21	28	33	38	58	82
B3. Combination of B1-B2 using one half standard deviation shocks	17	19	21	29	36	42	66	99
B4. One-time 30 percent real depreciation in 2015	17	29	33	38	43	47	63	88
B5. 10 percent of GDP increase in other debt-creating flows in 2015	17	30	36	42	47	52	69	90
PV of Debt-to-Revenue Ratio 2/								
Baseline	65	81	97	113	126	137	174	206
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	72	87	93	101	107	113	134	186
A2. Primary balance is unchanged from 2014	69	99	118	140	159	177	259	418
A3. Permanently lower GDP growth 1/	69	107	135	165	191	214	315	505
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviations in 2015-20	69	112	148	182	212	239	351	533
B2. Primary balance is at historical average minus one standard deviations in 2015-201	69	88	98	127	152	175	265	398
B3. Combination of B1-B2 using one half standard deviation shocks	69	87	95	129	159	186	298	475
B4. One-time 30 percent real depreciation in 2015	69	131	151	174	194	212	289	427
B5. 10 percent of GDP increase in other debt-creating flows in 2015	69	137	163	191	215	235	316	436
Debt Service-to-Revenue Ratio 2/								
Baseline	2	14	10	13	16	16	24	33
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	2	15	10	13	16	14	22	35
A2. Primary balance is unchanged from 2014	2	15	11	14	17	17	29	48
A3. Permanently lower GDP growth 1/	2	15	11	15	19	19	33	57
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviations in 2015-20	2	16	12	16	20	21	36	59
B2. Primary balance is at historical average minus one standard deviations in 2015-201	2	15	10	13	17	16	29	47
B3. Combination of B1-B2 using one half standard deviation shocks	2	16	10	13	17	17	32	54
B4. One-time 30 percent real depreciation in 2015	2	16	14	18	23	24	40	65
B5. 10 percent of GDP increase in other debt-creating flows in 2015	2	15	13	16	20	22	33	51

Sources: Haitian authorities; and staff estimates and projections.
1/ Assumes that real GDP growth is at baseline minus one standard deviation divided by the square root of the length of the projection period.
2/ Revenues are defined inclusive of grants.

Table 3a. Haiti: External Debt Sustainability Framework, Baseline Scenario, 2011-2034 1/

(In percent of GDP, unless otherwise indicated)

	Actual			Historical 6/ Average	Standard 6/ Deviation	Projections									
	2011	2012	2013			2014	2015	2016	2017	2018	2019	2014-2019 Average	2024	2034	2020-2034 Average
External debt (nominal) 1/	**8.9**	**13.8**	**17.7**			**20.1**	**24.6**	**28.7**	**32.3**	**35.7**	**38.6**		**48.8**	**53.3**	
o/w public and publicly guaranteed (PPG)	8.9	13.8	17.7			20.1	24.6	28.7	32.3	35.7	38.6		48.8	53.3	
Change in external debt	-4.0	4.9	3.9			2.4	4.5	4.1	3.7	3.3	3.0		1.4	-0.1	
Identified net debt-creating flows	0.7	1.9	4.2			3.5	3.3	3.0	2.5	2.0	1.5		0.4	-0.7	
Non-interest current account deficit	**3.4**	**5.3**	**6.4**	**2.7**	**2.3**	**5.7**	**5.5**	**5.4**	**5.2**	**5.0**	**4.7**		**4.4**	**4.7**	4.4
Deficit in balance of goods and services	40.7	36.3	34.2			32.0	31.1	30.2	29.3	28.3	27.4		23.6	16.3	
Exports	13.4	16.9	18.0			18.2	18.5	18.7	19.0	19.2	19.5		21.2	25.0	
Imports	54.2	53.2	52.2			50.3	49.6	48.9	48.2	47.6	46.9		44.7	41.3	
Net current transfers (negative = inflow)	-36.7	-35.9	-32.9	-31.6	6.7	-29.4	-28.6	-27.8	-26.9	-26.1	-25.4		-21.4	-13.0	-18.9
o/w official	-19.2	-12.5	-8.9			-7.8	-7.0	-6.7	-6.4	-6.2	-6.0		-4.8	-2.0	
Other current account flows (negative = net inflow)	-0.6	4.9	5.0			3.0	3.0	3.0	2.9	2.8	2.7		2.3	1.4	
Net FDI (negative = inflow)	**-2.4**	**-2.0**	**-1.4**	**-1.4**	**1.0**	**-1.7**	**-1.7**	**-1.8**	**-2.0**	**-2.1**	**-2.3**		**-3.0**	**-4.5**	-3.5
Endogenous debt dynamics 2/	**-0.3**	**-1.4**	**-0.8**			**-0.5**	**-0.5**	**-0.6**	**-0.7**	**-0.8**	**-0.9**		**-1.0**	**-0.9**	
Contribution from nominal interest rate	0.1	0.1	0.1			0.2	0.2	0.3	0.3	0.4	0.4		0.6	0.7	
Contribution from real GDP growth	-0.7	-0.2	-0.6			-0.7	-0.8	-0.9	-1.1	-1.2	-1.3		-1.6	-1.5	
Contribution from price and exchange rate changes	0.3	-1.3	-0.4			...	...	...	...	...	...		...	...	
Residual (3-4) 3/	**-4.8**	**3.0**	**-0.2**			**-1.1**	**1.2**	**1.1**	**1.2**	**1.3**	**1.4**		**1.0**	**0.6**	
o/w exceptional financing	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
PV of external debt 4/	...	...	11.4			13.6	16.6	19.4	21.9	24.3	26.4		34.1	38.2	
In percent of exports	...	...	63.1			74.4	89.9	103.5	115.6	126.3	135.7		161.3	153.0	
PV of PPG external debt	**...**	**...**	**11.4**			**13.6**	**16.6**	**19.4**	**21.9**	**24.3**	**26.4**		**34.1**	**38.2**	
In percent of exports	**...**	**...**	**63.1**			**74.4**	**89.9**	**103.5**	**115.6**	**126.3**	**135.7**		**161.3**	**153.0**	
In percent of government revenues	**...**	**...**	**89.3**			**102.6**	**123.0**	**140.8**	**156.0**	**169.2**	**180.2**		**213.8**	**210.6**	
Debt service-to-exports ratio (in percent)	**0.6**	**0.7**	**1.0**			**2.5**	**3.7**	**4.9**	**5.9**	**6.7**	**7.7**		**10.7**	**13.1**	
PPG debt service-to-exports ratio (in percent)	**0.6**	**0.7**	**1.0**			**2.5**	**3.7**	**4.9**	**5.9**	**6.7**	**7.7**		**10.7**	**13.1**	
PPG debt service-to-revenue ratio (in percent)	**0.8**	**0.9**	**1.4**			**3.4**	**5.0**	**6.6**	**7.9**	**9.0**	**10.2**		**14.2**	**18.0**	
Total gross financing need (Millions of U.S. dollars)	82.8	273.5	435.3			398.9	425.3	454.3	465.2	470.8	475.0		572.9	903.1	
Non-interest current account deficit that stabilizes debt ratio	7.5	0.4	2.4			3.2	1.0	1.4	1.6	1.7	1.8		3.0	4.8	
Key macroeconomic assumptions															
Real GDP growth (in percent)	5.5	2.9	4.3	1.5	3.4	4.0	4.0	4.0	4.0	4.0	4.0	4.0	3.5	3.0	3.3
GDP deflator in US dollar terms (change in percent)	-2.5	16.8	2.8	8.0	12.5	2.1	1.9	1.9	1.9	1.9	1.9	2.0	1.9	1.9	1.9
Effective interest rate (percent) 5/	0.5	0.8	0.9	1.1	0.7	1.0	1.3	1.3	1.2	1.3	1.3	1.2	1.3	1.3	1.3
Growth of exports of G&S (US dollar terms, in percent)	26.8	32.1	14.1	13.1	12.4	7.5	7.5	7.5	7.4	7.4	7.4	7.4	7.2	6.9	7.1
Growth of imports of G&S (US dollar terms, in percent)	-0.3	3.1	5.1	12.7	13.9	2.3	4.6	4.6	4.6	4.6	4.5	4.2	4.5	4.3	4.4
Grant element of new public sector borrowing (in percent)	...	...	...	...	...	37.5	36.7	36.5	36.4	36.3	36.2	36.6	36.1	35.9	36.0
Government revenues (excluding grants, in percent of GDP)	11.2	12.8	12.7			13.2	13.5	13.8	14.1	14.4	14.7		16.0	18.2	16.6
Aid flows (in Millions of US dollars) 7/	663.2	1211.5	1064.7			1058.7	1237.7	1283.5	1326.9	1384.0	1443.2		1626.3	1836.3	
o/w Grants	438.4	987.5	685.1			666.0	666.4	677.9	684.8	703.3	721.6		761.4	524.7	
o/w Concessional loans	224.8	224.0	379.6			392.7	571.2	605.6	642.0	680.7	721.6		864.9	1311.6	
Grant-equivalent financing (in percent of GDP) 8/	5.8	12.5	8.1			9.1	9.2	8.9	8.6	8.4	8.2		6.8	3.8	5.9
Grant-equivalent financing (in percent of external financing) 8/	100.0	100.0	100.0			76.8	70.8	70.0	69.2	68.7	68.1		66.0	54.2	62.5
Memorandum items:															
Nominal GDP (Millions of US dollars)	7516.0	7889.7	8458.4			8980.0	9520.6	10093.6	10700.7	11344.3	12026.7		15725.1	26232.9	
Nominal dollar GDP growth	13.5	5.0	7.2			6.2	6.0	6.0	6.0	6.0	6.0	6.0	5.5	5.0	5.3
PV of PPG external debt (in Millions of US dollars)			947.5			1200.1	1557.4	1927.5	2312.7	2715.6	3132.5		5285.3	9881.4	
(PVt-PVt-1)/GDPt-1 (in percent)						3.0	4.0	3.9	3.8	3.8	3.7	3.7	2.9	1.8	2.6
Gross workers' remittances (Millions of US dollars)	1375.6	1473.8	1552.7			1612.3	1781.0	1778.0	1885.1	1947.1	2009.7		2293.1	2650.6	
PV of PPG external debt (in percent of GDP + remittances)	...	...	9.6			11.5	14.0	16.5	18.6	20.7	22.6		29.8	34.7	
PV of PPG external debt (in percent of exports + remittances)	...	...	31.3			37.5	44.6	53.3	59.9	66.8	73.1		95.5	108.9	
Debt service of PPG external debt (in percent of exports + remittance	...	...	0.5			1.2	1.8	2.5	3.0	3.5	4.1		6.4	9.3	

Sources: Haitian authorities; and staff estimates and projections.

1/ Includes both public and private sector external debt.

2/ Derived as [r - g - ρ(1+g)]/(1+g+ρ+gρ) times previous period debt ratio, with r = nominal interest rate; g = real GDP growth rate, and ρ = growth rate of GDP deflator in U.S. dollar terms.

3/ Includes exceptional financing (i.e., changes in arrears and debt relief); changes in gross foreign assets; and valuation adjustments. For projections also includes contribution from price and exchange rate changes.

4/ Assumes that PV of private sector debt is equivalent to its face value.

5/ Current-year interest payments divided by previous period debt stock.

6/ Historical averages and standard deviations are generally derived over the past 10 years, subject to data availability.

7/ Defined as grants, concessional loans, and debt relief.

8/ Grant-equivalent financing includes grants provided directly to the government and through new borrowing (difference between the face value and the PV of new debt).

Table 3b.Haiti: Sensitivity Analysis for Key Indicators of Public and Publicly Guaranteed External Debt, 2014-2034
(In percent)

				Projections				
	2014	2015	2016	2017	2018	2019	**2024**	2034
PV of debt-to-GDP+remittances ratio								
Baseline	11	14	16	19	21	23	**30**	35
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2014-2034 1/	11	12	13	14	15	16	**20**	24
A2. New public sector loans on less favorable terms in 2014-2034 2	11	15	18	21	24	27	**39**	51
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2015-2016	11	14	18	20	23	25	**32**	38
B2. Export value growth at historical average minus one standard deviation in 2015-2016 3/	11	14	18	20	22	24	**30**	35
B3. US dollar GDP deflator at historical average minus one standard deviation in 2015-2016	11	15	18	20	23	25	**33**	38
B4. Net non-debt creating flows at historical average minus one standard deviation in 2015-2016 4/	11	17	22	23	25	27	**32**	35
B5. Combination of B1-B4 using one-half standard deviation shocks	11	16	21	23	25	26	**33**	37
B6. One-time 30 percent nominal depreciation relative to the baseline in 2015 5/	11	18	22	24	27	30	**39**	47
PV of debt-to-exports+remittances ratio								
Baseline	36	45	52	59	66	72	95	109
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2014-2034 1/	37	39	43	45	49	53	**72**	85
A2. New public sector loans on less favorable terms in 2014-2034 2	37	47	59	69	79	88	**124**	161
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2015-2016	37	44	53	59	66	72	**94**	107
B2. Export value growth at historical average minus one standard deviation in 2015-2016 3/	37	48	63	70	77	83	**105**	119
B3. US dollar GDP deflator at historical average minus one standard deviation in 2015-2016	37	44	53	59	66	72	**94**	107
B4. Net non-debt creating flows at historical average minus one standard deviation in 2015-2016 4/	37	56	73	75	81	86	**104**	110
B5. Combination of B1-B4 using one-half standard deviation shocks	37	49	62	66	72	78	**97**	107
B6. One-time 30 percent nominal depreciation relative to the baseline in 2015 5/	37	44	53	59	66	72	**94**	107
PV of debt-to-revenue ratio								
Baseline	103	123	141	156	169	180	**214**	211
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2014-2034 1/	103	106	109	113	117	122	**141**	138
A2. New public sector loans on less favorable terms in 2014-2034 2	103	130	156	179	199	217	**278**	311
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2015-2016	103	129	156	173	187	200	**237**	233
B2. Export value growth at historical average minus one standard deviation in 2015-2016 3/	103	127	155	169	182	191	**219**	209
B3. US dollar GDP deflator at historical average minus one standard deviation in 2015-2016	103	129	158	175	190	202	**240**	236
B4. Net non-debt creating flows at historical average minus one standard deviation in 2015-2016 4/	103	145	182	196	206	213	**232**	213
B5. Combination of B1-B4 using one-half standard deviation shocks	103	139	179	194	207	216	**243**	229
B6. One-time 30 percent nominal depreciation relative to the baseline in 2015 5/	103	172	197	218	236	251	**298**	294

Table 3b. Haiti: Sensitivity Analysis for Key Indicators of Public and Publicly Guaranteed External Debt, 2014-2034 (concluded)

(In percent)

Debt service-to-exports+remittances ratio								
Baseline	1	2	3	3	4	4	**6**	9
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2014-2034 1/	1	2	2	3	3	3	**4**	6
A2. New public sector loans on less favorable terms in 2014-2034 2	1	2	3	4	5	6	**9**	15
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2015-2016	1	2	3	3	4	4	**6**	9
B2. Export value growth at historical average minus one standard deviation in 2015-2016 3/	1	2	3	3	4	5	**7**	10
B3. US dollar GDP deflator at historical average minus one standard deviation in 2015-2016	1	2	3	3	4	4	**6**	9
B4. Net non-debt creating flows at historical average minus one standard deviation in 2015-2016 4/	1	2	3	3	4	5	**7**	10
B5. Combination of B1-B4 using one-half standard deviation shocks	1	2	3	3	4	5	**7**	9
B6. One-time 30 percent nominal depreciation relative to the baseline in 2015 5/	1	2	3	3	4	4	**6**	9
Debt service-to-revenue ratio								
Baseline	3	5	7	8	9	10	**14**	18
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2014-2034 1/	3	5	6	7	7	7	**9**	10
A2. New public sector loans on less favorable terms in 2014-2034 2	3	5	7	10	12	14	**21**	29
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2015-2016	3	5	7	9	10	11	**16**	20
B2. Export value growth at historical average minus one standard deviation in 2015-2016 3/	3	5	7	8	10	11	**15**	18
B3. US dollar GDP deflator at historical average minus one standard deviation in 2015-2016	3	5	8	9	10	12	**16**	21
B4. Net non-debt creating flows at historical average minus one standard deviation in 2015-2016 4/	3	5	7	9	11	13	**16**	19
B5. Combination of B1-B4 using one-half standard deviation shocks	3	5	7	9	11	13	**17**	20
B6. One-time 30 percent nominal depreciation relative to the baseline in 2015 5/	3	7	9	11	13	14	**20**	26
Memorandum item:								
Grant element assumed on residual financing (i.e., financing required above baseline) 6/	34	34	34	34	34	34	**34**	34

Sources: Haitian authorities; and staff estimates and projections.

1/ Variables include real GDP growth, growth of GDP deflator (in U.S. dollar terms), non-interest current account in percent of GDP, and non-debt creating flows.

2/ Assumes that the interest rate on new borrowing is by 2 percentage points higher than in the baseline., while grace and maturity periods are the same as in the baseline.

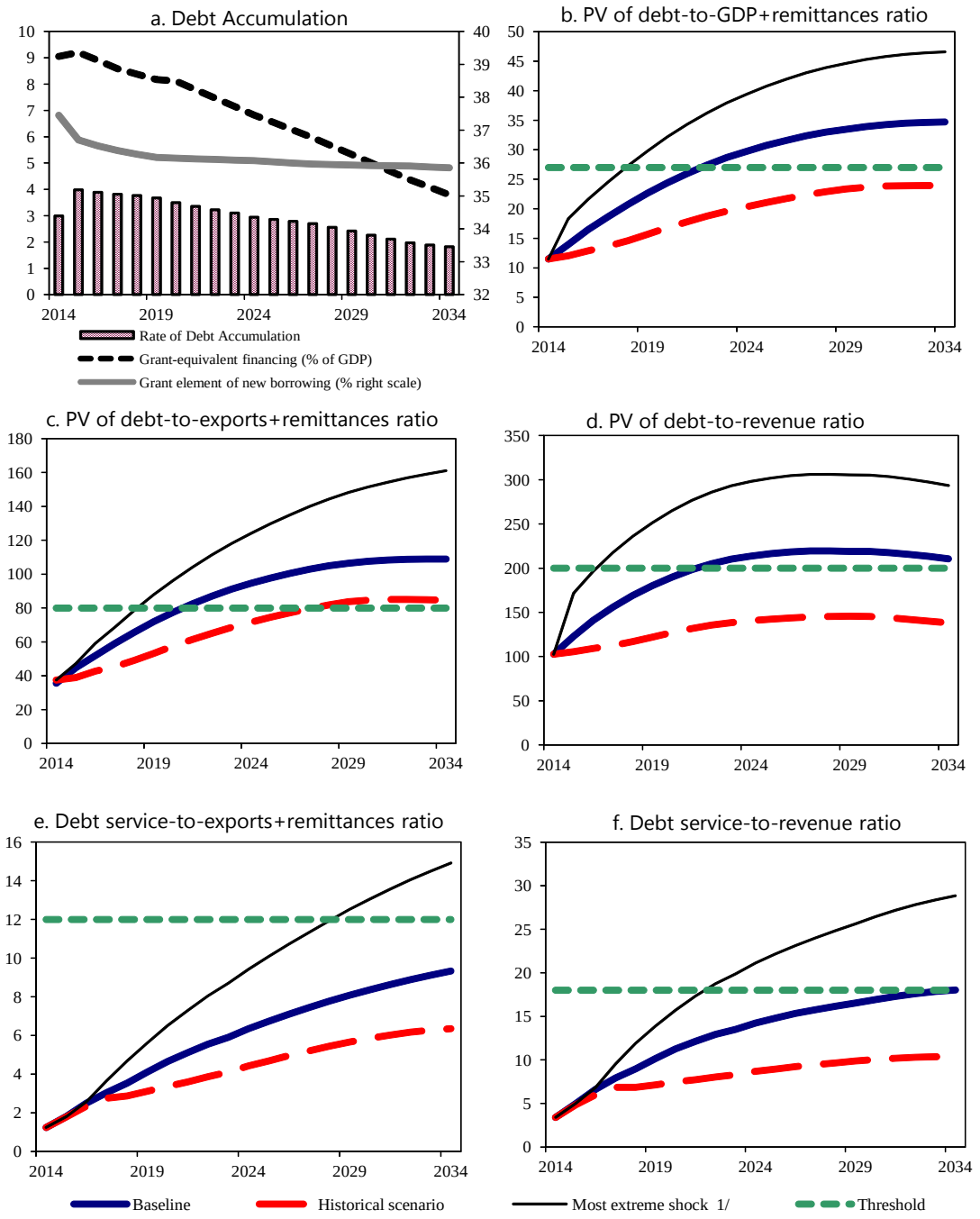
3/ Exports values are assumed to remain permanently at the lower level, but the current account as a share of GDP is assumed to return to its baseline level after the shock (implicitly assuming an offsetting adjustment in import levels).

4/ Includes official and private transfers and FDI.

5/ Depreciation is defined as percentage decline in dollar/local currency rate, such that it never exceeds 100 percent.

6/ Applies to all stress scenarios except for A2 (less favorable financing) in which the terms on all new financing are as specified in footnote 2.

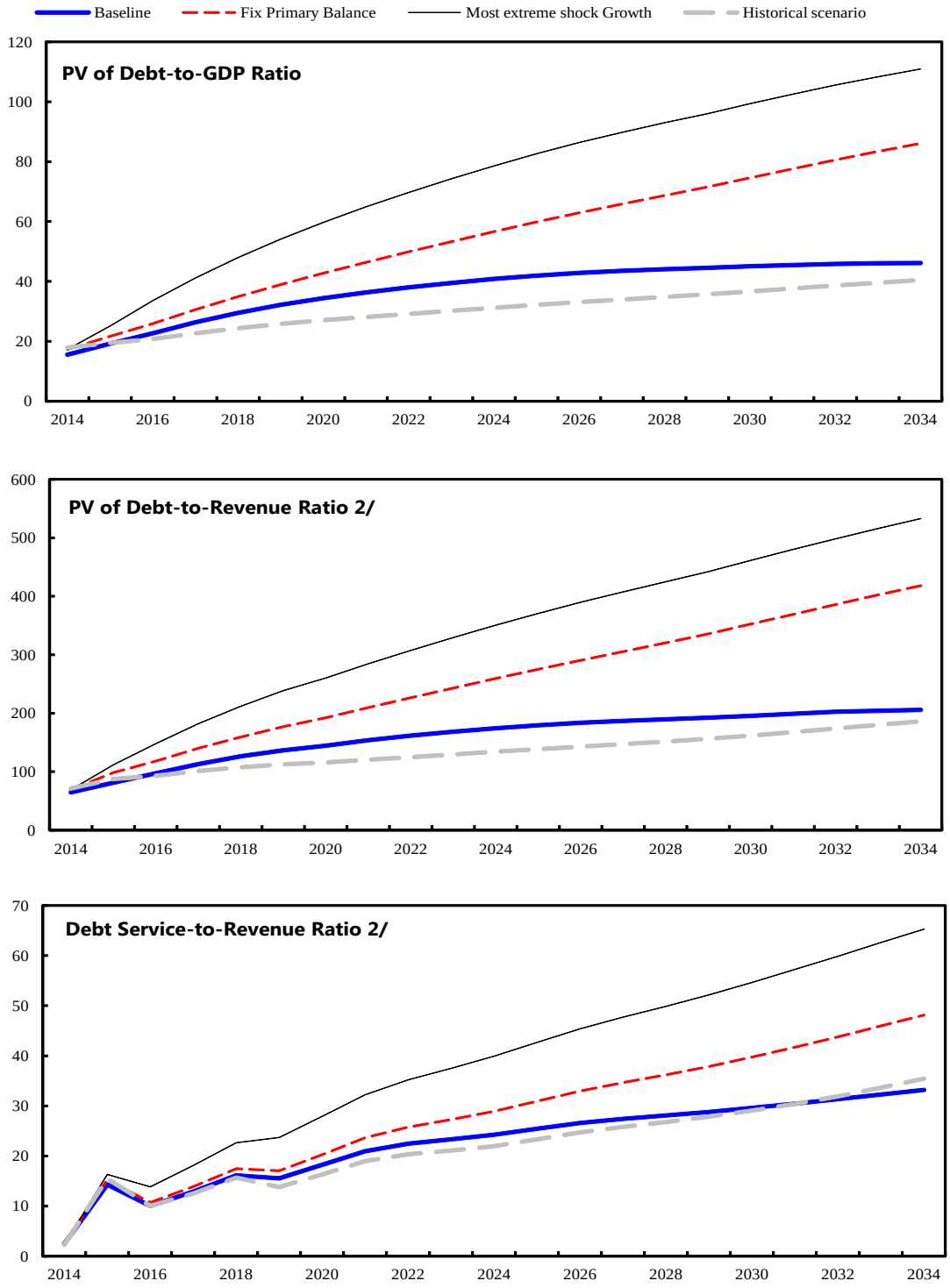
Figure 1. Haiti: Indicators of Public and Publicly Guaranteed External Debt under Alternatives Scenarios, 2014-2034 [1/]



Sources: Haitian authorities; and staff estimates and projections.

1/ The most extreme stress test is the test that yields the highest ratio in 2024. In figure b. it corresponds to a One-time depreciation shock; in c. to a Terms shock; in d. to a One-time depreciation shock; in e. to a Terms shock and in figure f. to a Terms shock

Figure 2. Haiti: Indicators of Public Debt Under Alternative Scenarios, 2014-2034 1/



Sources: Haitian authorities; and staff estimates and projections.
1/ The most extreme stress test is the test that yields the highest ratio in 2024.
2/ Revenues are defined inclusive of grants.